Cellular Immunology 287 (2014) 62–68

Contents lists available at ScienceDirect

Cellular Immunology

journal homepage: www.elsevier.com/locate/ycimm

Trikatu, an herbal compound as immunomodulatory and anti-inflammatory agent in the treatment of rheumatoid arthritis – An experimental study

Vachana Murunikkara, MahaboobKhan Rasool *

Immunopathology Lab, School of Bio Sciences and Technology, VIT University, Vellore 632 014, Tamil Nadu, India

ARTICLE INFO

Article history:
Received 4 September 2013
Accepted 11 December 2013
Available online 18 December 2013

Keywords:
Trikatu
Immune response
Macrophage phagocytic index
Inflammatory mediators
Autoimmune disease

ABSTRACT

In the present study, trikatu, an herbal compound was evaluated for its immunomodulatory and anti-inflammatory properties with reference to cell mediated immune responses (delayed type hypersensitivity reaction), humoral immune response (haemagglutination titer and plaque forming assay), macrophage phagocytic index, circulating immune complex and inflammatory mediators in rats. For comparison purposes, indomethacin was used as a reference drug for anti-inflammatory studies. The results obtained in our study showed a significant decrease in cell mediated immune responses, humoral immune responses (haemagglutination titre and plaque forming assay) and macrophage phagocytic index in trikatu treated rats (1000 mg/kg/b.wt.) compared to control animals implying its immunosuppressive property. In addition, significant anti-inflammatory effects were observed in trikatu treated adjuvant induced arthritic rats by a reduction in the levels of circulating immune complexes and inflammatory mediators (TNF-alpha and Interleukin-1beta). Thus, in conclusion, our data suggest that trikatu could be considered as a potential anti-inflammatory agent for treating autoimmune inflammatory disorders like rheumatoid arthritis with immunosuppressive property.

1. Introduction

Rheumatoid arthritis is a chronic systemic autoimmune inflammatory disorder of unknown origin. It is characterized by the inflammation of synovial joints with neovascularization and pannus formation, the expanded destructive synovial tissue responsible for cartilage and bone erosion in RA. Although the etiology is unknown, macrophages, B cells, mast cells, synovial fibroblast and CD4+ T cells become activated during rheumatoid arthritis and act as main culprits to cause joint inflammation and destruction [1]. Moreover, the pro-inflammatory cytokines appear to play a central role in the development of pathogenesis of rheumatoid arthritis. The increased flow of TNF-α, IL-1β, IL-6, MCP-1, VEGF and PGE_2 from the activated immune cells into the synovium during rheumatoid arthritis are found to mediate synovial joint inflammation [2]. Thus, it has been hypothesized that modulating the activation of immune cells and suppression of pro-inflammatory cytokine levels could serve as a basis for the potential development of anti-inflammatory drug with immunomodulatory activity.

Immunomodulation is the process of modifying an immune response in a positive or negative manner by the administration of a drug or compound. Apart from being specifically stimulatory or suppressive, certain agents normalize or modulate pathophysiological processes and hence are called immunomodulatory agents [3]. Trikatu, an herbal compound is made up of by three crude drugs namely dried fruits of black pepper (*Piper nigrum* Linn.), dried fruits of long pepper (*Piper longum* Linn.) and dried rhizomes of ginger (*Zingiber officinalis* Rosc.) in the ratio of (1:1:1; w:w) [4]. Trikatu has been used as a essential ingredient of many herbal formulation and used traditionally in the treatment of gastric and abdominal disorders, asthma, bronchitis, coughs, dysentery, pyrexia, insomnia, colic and intestinal infection. Trikatu, significantly increase the bioavailability of other compounds, including herbal, nutrients and pharmaceutical drugs. Recent reports have also indicated the anti-inflammatory effect of *Z. officinalis* Rosc. against adjuvant-induced arthritis [5]. Our preliminary studies with the trikatu and its active principles like piperine, 6-shogaol and 6-gingerol showed promising anti-inflammatory effect in both rheumatoid arthritis and acute gouty arthritis [6,7]. Moreover, the development and pathogenesis of inflammation and autoimmune diseases are deeply correlated with immunological function status. Thus, the encouraging results of trikatu and its components observed from our and other studies against rheumatoid arthritis paved the way to explore its potential immunomodulatory and anti-inflammatory effects in experimental animals against some

* Corresponding author.
E-mail address: mkr474@gmail.com (M. Rasool).

0008-8749/$ - see front matter © 2013 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.cellimm.2013.12.002

dysregulated immune condition which is closely observed in rheumatoid arthritis.

2. Materials and methods

2.1. Animals

Female albino rats of Wistar strain, 130 ± 20 g, were procured from the Animal House, VIT University, Vellore, India. They were acclimatized for a week in a light and temperature-controlled room with a 12-h dark–light cycle and fed with commercial pelleted feed from Hindustan Lever Ltd. (Mumbai, India) and water was freely available. The animals were treated and cared for in accordance with the guidelines recommended by the Committee for the Purpose of Control and Supervision of Experiments on Animals, Government of India. The experimental protocol was approved by our departmental ethics committee.

2.2. Drug

Black pepper (dried fruit of *P. nigrum* Linn.), long pepper (dried fruit of *P. longum* Linn.) and ginger (dried rhizome of *Z. officinale* Rosc.) were procured from herbal drug supplier, Chennai, India and authenticated by chief botanist, VIT University, India. The laboratory trikatu was prepared by mixing the three powdered fruits in equal proportion according to Ayurvedic formulary. Indomethacin was purchased from Tamil Nadu Dadha Pharmaceuticals Ltd., Chennai, India. A homogenous aqueous suspension of trikatu and indomethacin was made with 0.5% carboxy methyl cellulose in phosphate buffered saline. Fresh solution was prepared before each experiment. All other reagents used were standard laboratory reagents of analytical grade and were purchased locally.

2.3. Dosage

Based on our preliminary studies with different dosages (500, 750 and 1000 mg/kg/b.wt.) of this trikatu, it was found that 1000 mg/kg/b.wt. oral dosage produced significant anti-inflammatory effect by reducing paw swelling in monosodium urate crystal-induced rats. Hence 1000 mg/kg/b.wt. dosage was considered for this study. The dosage of standard drug indomethacin (3 mg/kg/b.wt.) used in this study was selected based on our previous report [8].

2.4. HPLC profiling of trikatu

HPLC profiling of trikatu was performed with Shimadzu liquid chromatography system equipped with a model LC-10 AT VP solvent delivery pump, a Hamilton injector (loop volume of 20 μl) and a SPD-10AV variable wavelength UV–vis detector. The samples were separated at 27 °C on a minibore Phenomenex Luna 5-μm C_{18} column, dimensions 250 × 4.60 mm, using isocratic elution. The mobile phase comprised methanol and water in the ratio 68:32. The flow rate was 1 ml/min. The HPLC was monitored at 340, 280 and 200 nm for piperine, 6-gingerol and vitamin C respectively. Data was analyzed using LC real time software.

2.5. Immunomodulatory activity of trikatu

2.5.1. Delayed type hypersensitivity response (DTH)

The effect of trikatu on the antigen specific cellular immune response in experimental animals was measured by determining the degree of DTH response using the foot paw swelling test [9]. Rats were injected intraperitoneally with a suspension containing 1×10^6 sheep red blood cells (SRBC) in 0.2 ml of phosphate buffered saline (PBS) on day zero sensitization. Seven days later (day +7), the same animals were injected subcutaneously with 1×10^6 SRBC suspended in 50 μl of PBS into the right hind foot pad for elicitation of the DTH reaction. The left hind foot pad was injected with 50 μl of PBS as a control. Foot pad swelling was measured on day +8 with a caliper. The difference between the means of right and left hind foot-pad thickness gave a degree of foot pad swelling which was used for group comparisons. To establish the effect of trikatu on this immune response, a daily dose of trikatu (1000 mg/kg/b.wt.) was administered intraperitoneally at the induction phase (+4 to +7 days). Simultaneously, another group of animals (controls) was administered in the same condition with 0.1 ml of PBS. Each experimental group contains 6 animals.

2.5.2. Humoral immune response in rats

Rats were intraperitoneally immunized with 0.2 ml, 5×10^9 cells/ml of sheep red blood cells (SRBC) on day zero. Trikatu (1000 mg/kg/b.wt.) was administered orally on days −3, −2, −1, 0,+1,+2 and +3. On day 7 after immunization animals the rats were sacrificed by euthanasia. Blood was collected for serum separation and haemagglutination assay was performed to determine the antibody titre. The spleen was dissected out and the plaque forming assay was performed and the number of plaque forming cells were determined. Each experimental group contains 6 animals.

2.5.2.1. Haemagglutination assay. Haemagglutination assay was performed by the method of Subramoniam et al. [10]. Serum obtained was diluted two fold using 0.15 M phosphate buffered saline (pH 7.2) and 50 μl of each dilution was aliquoted into 96-well plate microtiter plates. A 25 μl of fresh 1% SRBC suspension in the above buffer was dispersed into each well and mixed. The plates were incubated at 37 °C for 2 h and examined visually for agglutination. The value of the highest serum dilution causing haemagglutination was taken as antibody titre. Antibody titre was expressed in a graded manner, the minimum dilution (1/2) being ranked as 1, and the mean ranks of different groups were compared for statistical significance.

2.5.2.2. Plaque forming assay. Plaque forming assay was performed by the method of Jerne and Nordine [11]. The spleen cells were separated and collected in RPMI-1640 medium, washed twice and suspended in the same medium to a concentration of 10^6 cells/ml. Glass petri dishes were layered with 1.2% agarose in 0.15 M NaCl to form a bottom layer of agarose and incubated at 37 °C for 90 min. A 2 ml quantity of 1:9 diluted fresh rabbit serum was added to each petri dish and further incubated for 4 min to allow the formation of plaques. Plaques were counted immediately using an inverted microscope (Nikon H-III power) and the values were expressed as counts per 10^6 spleen cells. The values were compared for statistical significance.

2.5.3. Estimation of macrophage phagocytic index

Trikatu was injected intraperitoneally (100 mg/kg/b.wt.) for 5 consecutive days. Control rats were given an equal volume of phosphate buffered saline (pH 7.4). After 48 h of last dose, rats were injected via the tail vein with colloidal carbon (India ink), which was diluted with PBS (pH 7.4) to eight times before, use (10 μl/g body weight.). Blood samples were drawn from the retro-orbital plexus at 0 and 15 min. The blood (25 μl) was dissolved in 0.1% sodium carbonate (2 ml) and the absorbance was measured at 660 nm [12]. The phagocytic index, *K*, was calculated by the equation:

$$K = (\ln \mathrm{OD}_1 - \ln \mathrm{OD}_2)/(t_2 - t_1) \quad (1)$$

where, OD_1 and OD_2 depict the optical densities at times t_1 and t_2, respectively.

2.5.4. Anti-inflammatory activity of trikatu

2.5.4.1. Induction of arthritis. The rats were divided into four groups each comprising six animals. Group 1 served as controls. In group 2, arthritis was induced by a single intradermal injection of Complete Freund's Adjuvant (0.1 ml) into the foot pad of right hind paw [13]. The adjuvant contained heat-killed *Mycobacterium tuberculosis* (10 mg) in paraffin oil (1 ml). Group 3 and 4 comprised of arthritic rats treated orally and intraperitoneally with trikatu (1000 mg/kg/b.wt.) and indomethacin (3 mg/kg/b.wt.) respectively, for 8 days from day 11 to 18 post adjuvant administration. The development of arthritis was monitored by measuring the paw thickness using a vernier scale periodically. On day 19, at the end of experimental period, the animals were sacrificed and blood was collected for serum separation. The serum was used for estimating circulating immune complex and cytokine levels.

2.5.4.2. Estimation of circulating immune complex in sera. The circulating immune complex levels in the serum of control and experimental animals were measured by the method of Seth and Srinivas [14]. Briefly, 2 ml of 4.166% polyethylene glycol (PEG) 6000 in 0.1 M borate buffered saline (BBS), pH 8.4 was added to 0.22 ml of sera diluted 1:3 in BBS to obtain a final concentration of 3.75% PEG and 1:30 of serum. A similar volume of diluted serum mixed with 2 ml of BBS served as a control. After incubation at 25 °C for 60 min, the light absorbance of the test and control samples was measured at 450 nm (E_{450}) using a spectrophotometer. The results are expressed as PEG index derived by the following formula:

$$\text{PEG index} = (E_{450} \text{ with PEG} - E_{450} \text{ with BBS}) \times 1000 \qquad (2)$$

The PEG method for estimating CICs was preferred because the method is simple, less time consuming and equally sensitive to the complement consumption assay.

2.5.4.3. Estimation of cytokines. Proinflammatory cytokines like IL-1 β and TNF-α were measured in the serum of control and experimental animals using enzyme linked immunosorbent assay (ELISA) kits according to the procedure recommended by the manufacturer (PeproTech, USA) instructions.

2.6. Statistical analysis

Results were expressed as mean ± SD and statistical analysis was performed by GraphPad InStat 3 software, using ANOVA to determine the significant differences between groups, followed by Student's Newman–Keul's test. $p < 0.05$ implied significance.

3. Results

3.1. HPLC analysis of trikatu

Fig. 1 shows the HPLC profile of trikatu, which was recorded at 340, 280 and 200 nm. On comparison with standards, identical retention time confirms the presence of piperine (peak retention time (t_R) 13.529 min), 6-gingerols (peak retention time (t_R) 9.469 min) and vitamin C (peak retention time (t_R) 2.964 min) in trikatu.

3.2. Effect of trikatu on delayed type hypersensitivity response

Inorder to assess the trikatu's immunosuppressive effect. The cell mediated immune response against sheep red blood cell antigen was determined by using delayed type hypersensitivity reaction test. The difference in mean foot pad thickness between the right and left hind limbs at the end of the experimental period is expressed as the DTH edema index. Animals treated with trikatu (1000 mg/kg/b.wt.) showed a significant ($p < 0.05$) reduction in inflammation due to the DTH response compared to the untreated rats, as measured by a decreased DTH edema index (Fig. 2).

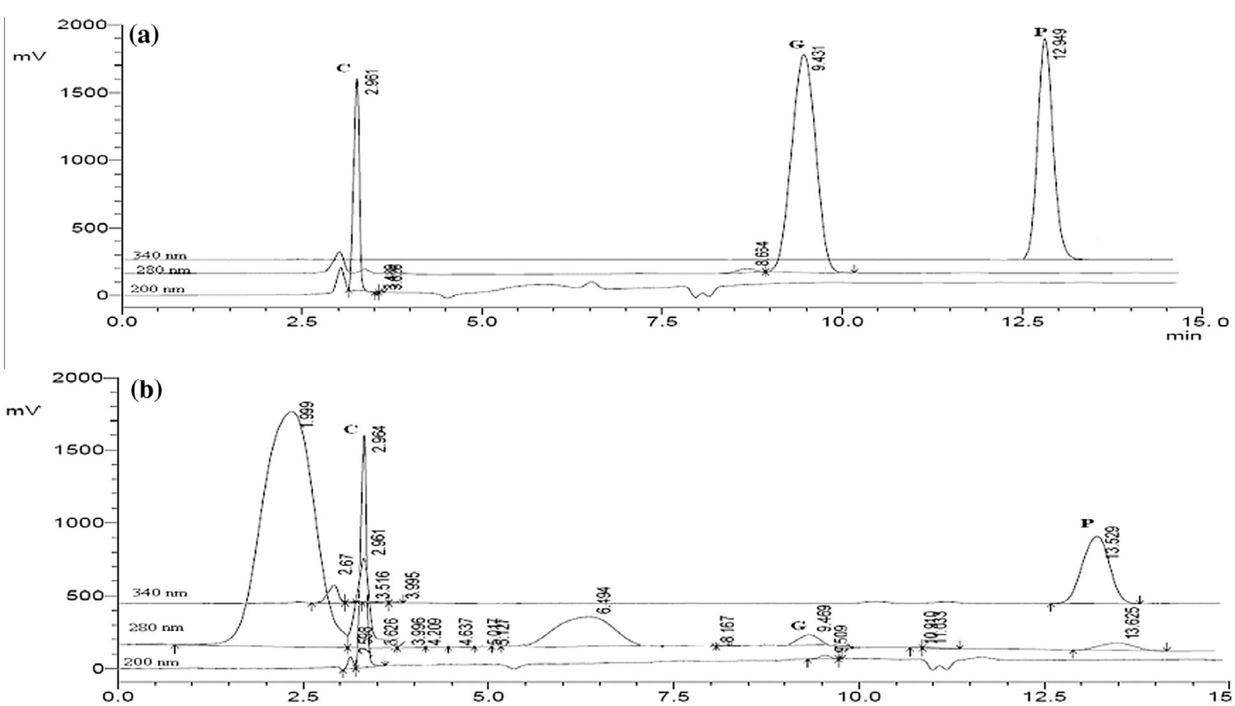

Fig. 1. HPLC chromatograms of standard and trikatu. (a) Shows the HPLC chromatograms of standard run at 200, 280 and 340 nm for vitamin C, 6-gingerols and piperine, respectively. Peaks (C) vitamin C, (G) 6-gingerol, (P) piperine. (b) Shows the HPLC chromatograms of trikatu run at 200, 280 and 340 nm. Peaks (C) vitamin C, (G) 6-gingerol, (P) piperine with identical retention time as that of standard.

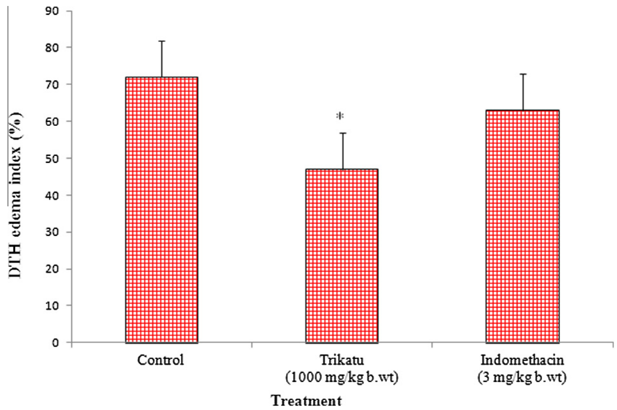

Fig. 2. Effect of trikatu and indomethacin on delayed type hypersensitivity response in rats. Administration of trikatu (oral) and indomethacin (i.p.) were done on days +4 to +7 post-sensitization with SRBC on day 0. On day +7, subcutaneous injection of SRBC in the right hind foot pad of rats elicited DTH response. Difference in mean foot pad thickness between the right and left hind limbs at the end of experimental period was expressed as DTH edema index. Trikatu and indomethacin treatment reduced the DTH edema index. Bars are expressed as mean ± SD of six animals; Comparisons are made with control group. The symbols represent statistical significance at $^{*}p < 0.05$.

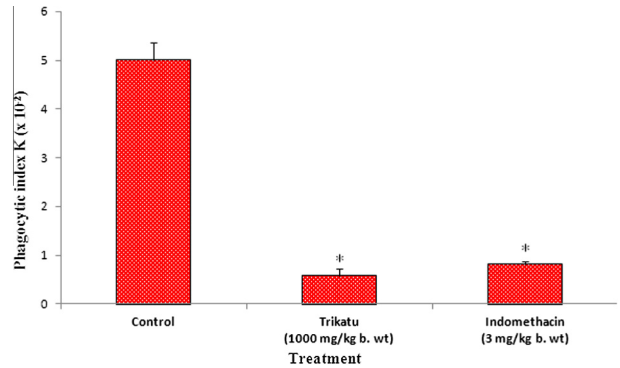

Fig. 4. Effect of trikatu and indomethacin on macrophage phagocytic index in rats. Trikatu and indomethacin were administered orally and i.p., respectively for five consecutive days. The colloidal carbon suspension was injected i.v. 48 h after the last drug dose. The phagocytic activity was expressed as the phagocytic index *K*. Treatment with trikatu and indomethacin showed a significant reduction in phagocytic index. Bars are expressed as mean ± SD of six animals; Comparison was made with control group. The symbols represent statistical significance at $^{*}p < 0.05$.

3.3. Effect of trikatu on humoral immune response

To appraise the effect of trikatu on humoral immune response (Fig. 3); haemagglutination assay and plaque forming assay were carried out. A significant decrease in the antibody titre/No. of plaque forming cells was observed in trikatu (1000 mg/kg/b.wt.) administered rats when compared to control group indicating the immunosuppressive role of trikatu.

3.4. Effect of trikatu on macrophage phagocytic index

The in vivo macrophage phagocytic capacity of the reticuloendothelial system was determined to evaluate the immunomodulatory effect of trikatu by the carbon clearance method. The phagocytic index *K* gives a measure of the rate of colloidal carbon clearance from circulation by these phagocytic cells. Treatment with trikatu (1000 mg/kg/b.wt.) significantly ($p < 0.05$) lowered the macrophage phagocytic index in rats when compared to the untreated control rats (Fig. 4). This indicates suppression in macrophage function by trikatu.

3.5. Effect of trikatu on circulating immune complex levels (CIC) in serum

The circulating immune complexes formed between autoantibodies and self-antigens in adjuvant-induced arthritis are likely to have a role in disease pathology. The polyethylene glycol (PEG) method employs a turbidimetric assay using small concentrations of PEG 6000 which precipitates complexed (but not free) immunoglobulins. The effect of trikatu on the levels of CIC in the sera of control and experimental arthritic rats is shown in Fig. 5. An increase in CIC levels was found in the arthritic rats when compared to the control rats. Administration of trikatu (1000 mg/kg/b.wt.) significantly ($p < 0.05$) reduced the CIC levels in arthritic rats comparable to indomethacin treatment.

3.6. Effect of trikatu on cytokine analysis in serum

Fig. 6 depict the levels of TNF-α and IL-1β respectively in the serum of control and experimental rats. A significant upregulated levels of TNF-α and IL-1β were observed in arthritic rats while trikatu treated rats showed a significant ($p < 0.05$) decrease in the levels of pro-inflammatory cytokines (TNF-α and IL-1β).

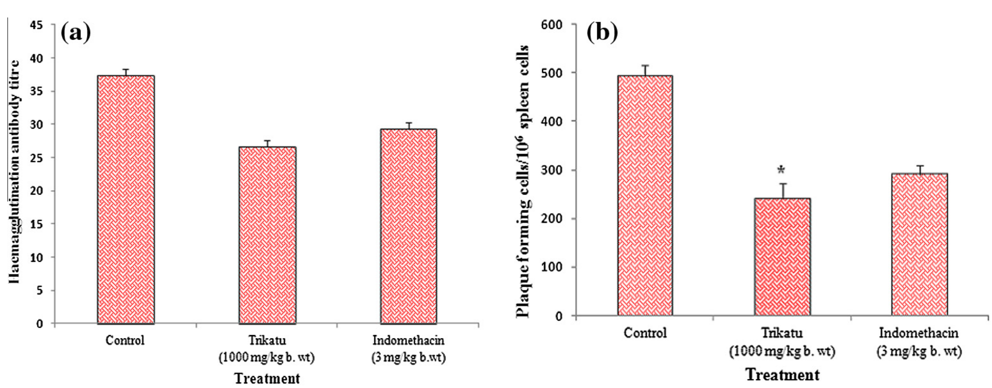

Fig. 3. Effect of trikatu and indomethacin on humoral immune response. (a) and (b) Shows the haemagglutination antibody titre and plaque formation respectively of trikatu and indomethacin. Administration of trikatu (oral) and indomethacin (i.p.) were done on days −3 to +3, pre and post-sensitization with SRBC on day 0. Values are expressed as mean ± SD of six animals. Comparison was made with control group. The symbols represent statistical significance at $^{*}p < 0.05$.

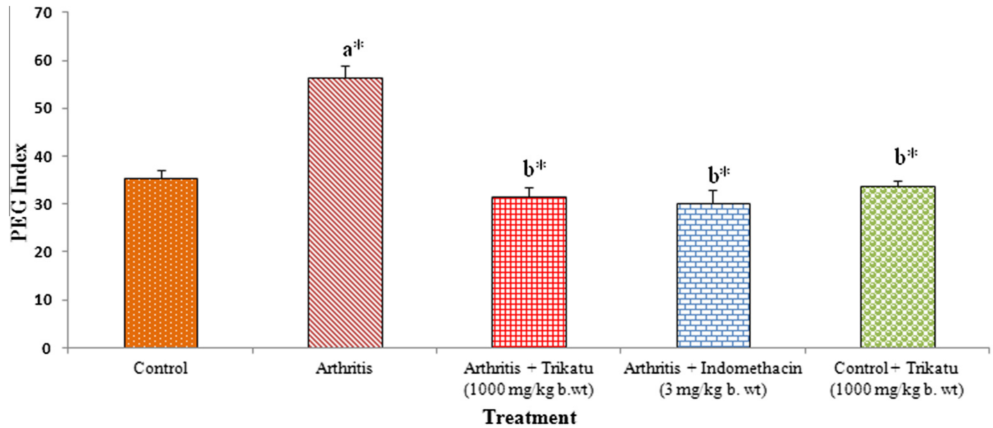

Fig. 5. Effect of trikatu on circulating immune complex levels in adjuvant-induced arthritic rats. Arthritis was induced in rats by a single intra-dermal foot pad injection of Complete Freund's Adjuvant into right hind paw. The test drugs were administered orally (trikatu) and i.p. (indomethacin) from days 11 to 18 post-adjuvant administration. On the 19th day, blood was collected and serum was used for estimating CIC levels using polyethylene–glycol-based turbidometric assay. The levels of CIC in serum were expressed as the PEG index. Treatment with trikatu significantly reduced the serum CIC levels when compared to untreated arthritic rats. Bars are expressed as mean ± SD of six animals. Comparisons are made as follows: (a) Control rats vs arthritic rats, trikatu-treated arthritic rats, indomethacin-treated arthritic rats and trikatu-treated control rats. (b) Arthritic rats vs trikatu-treated arthritic rats, indomethacin-treated arthritic rats and trikatu-treated control rats. The symbol represents statistical significance at $^{*}p < 0.05$.

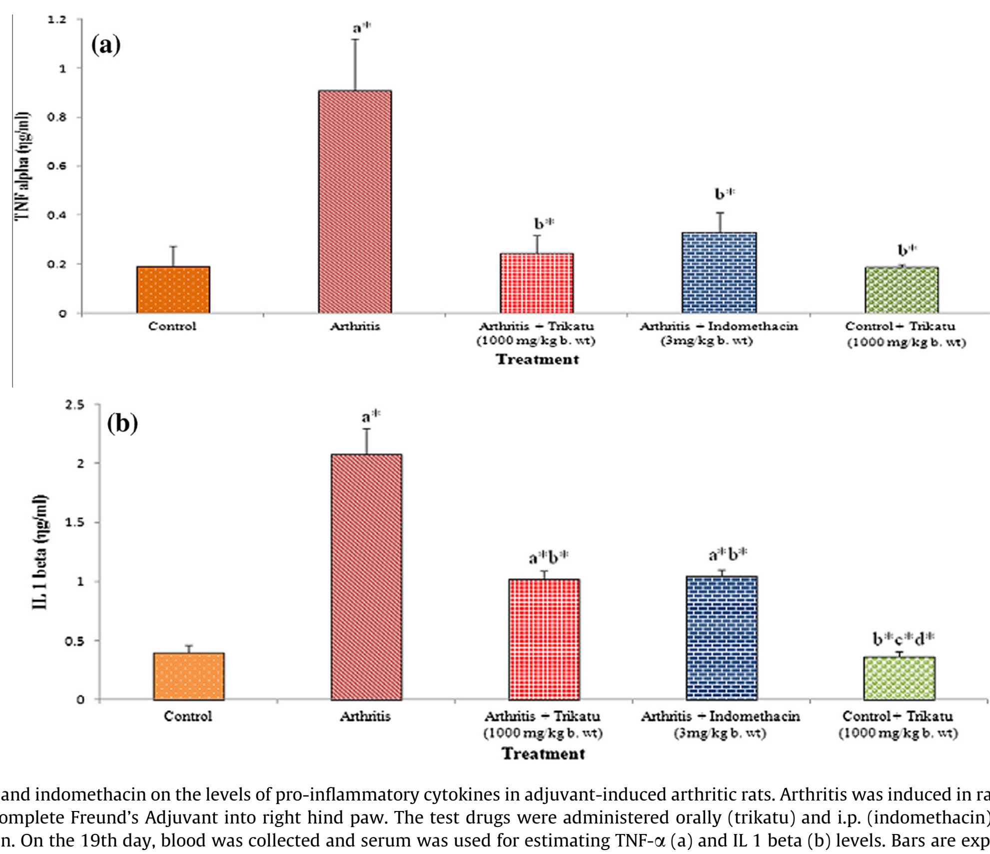

Fig. 6. Effect of trikatu and indomethacin on the levels of pro-inflammatory cytokines in adjuvant-induced arthritic rats. Arthritis was induced in rats by a single intra-dermal foot pad injection of Complete Freund's Adjuvant into right hind paw. The test drugs were administered orally (trikatu) and i.p. (indomethacin) from days 11 to 18 post-adjuvant administration. On the 19th day, blood was collected and serum was used for estimating TNF-α (a) and IL 1 beta (b) levels. Bars are expressed as mean ± SD of six animals. Comparisons are made as follows: (a) Control rats vs arthritic rats, trikatu-treated arthritic rats, indomethacin-treated arthritic rats and trikatu-treated control rats. (b) Arthritic rats vs trikatu-treated arthritic rats, indomethacin-treated arthritic rats and trikatu-treated control rats. The symbol represents statistical significance at $^{*}p < 0.05$.

4. Discussion

The involvement of immune system is considered as one of the main factors in the etiology and pathophysiology of rheumatoid arthritis and it is the mostly widely accepted hypothesis. Usually, the immunomodulatory drugs used for the treatment of autoimmune disorders has the ability to suppress the undesired immune function. Hence, in the present study, we evaluated the effect of trikatu on some immunological parameters like cell mediated immune response, humoral immune response, macrophage phagocytic index, circulatory immune complex and cytokine levels which are all closely related to rheumatoid arthritis in order to examine it immunomodulatory and anti-inflammatory effect. This investigation provided us an evidence that trikatu possesses immunomodulatory and anti-inflammatory effect.

The measurement of delayed type hypersensitivity reaction to commonly encountered antigens is widely accepted method to assess the cell mediated immune response. DTH response is an essential component of the evaluation of the immune system. A delayed type hypersensitivity reaction is an immune-inflammatory response involving introduction of an antigen to which an individual has been previously exposed and to assess the capacity of memory T-lymphocytes to respond to that antigen [15]. Upon same antigen presentation, sensitized T cells secrete various cytokines that induce Th1 cell proliferation and recruit macrophages into the inflamed area, promoting secretion of IL-12, TNF-α and IL-1 [16]. Humoral immune response, via antibody response involves the interaction of B cells with the antigen and their subsequent proliferation and differentiation into antibody secreting cells which secrete antibodies. In the present study, SRBC-induced delayed type hypersensitivity (DTH) was used to assess the effect of trikatu on cell mediated immunity and haemagglutination antibody titer and plaque forming assay were used to assess the humoral immune response. From the results obtained in our study, the administration of trikatu produced a significant reduction in both DTH and humoral immune response (antibody titer, and number of plaque) compared to untreated rats. This inhibitory action observed could be due to an influence of ginger and pipernic acid (active principle of *P. longum*) present in trikatu which has been already well established [17,18]. This confirms the immunosuppressive effect of the drug.

Phagocytosis is the process by which cells recognize and engulf large particles and it is important for host defense mechanisms as well as for tissue repair and morphogenetic remodeling. Macrophages are potent phagocytic cells in many inflammatory responses. They are professional antigen-presenting cells activating T cells through antigen presentation and co-stimulatory molecules [19]. Activated macrophages subsequently produce a variety of proinflammatory cytokines and chemokines to promote the development of inflammation and inflammatory disorders like arthritis. Therefore, subtle counteraction of macrophage activation has been proven beneficial in suppressing inflammation. In this study, the effect of trikatu on the phagocytic activity of the reticulo-endothelial system was evaluated using the carbon clearance test in rats. The carbon clearance was done to evaluate the effect of drugs on the phagocytic activity of the macrophage phagocyte system. The reticulo-endothelial system consists of phagocytic cells (macrophages); play an important role in the clearance of the particle from the blood stream. Normally, the intravenously injected colloidal carbon particles are cleared from the circulation in an hour [20]. In our present study, trikatu treatment produced a significantly reduced phagocytic responsiveness as observed by a decreased rate of carbon clearance compared to the untreated control rats. This proves the immunosuppressive effect of the drug on the macrophage phagocytic system.

Rheumatoid arthritis is linked to a high level of circulatory immune complex levels in synovial fluid. Normally, immune complexes circulate in the blood, synovial fluid, or lymphatic fluid until they are consumed by immune cells called macrophages. With rheumatic diseases, however, immune complexes attract immune system proteins called "complement". Complement proteins attach to this immune complex attract other immune cells that engender chemical attacks and consequent tissue damage [21,22]. In accordance to this concept, in our present study, the circulatory immune complex levels were found increased in the serum of adjuvant-induced arthritic rats when compared to the control rats. However, suppression in the circulatory immune complex levels were observed in the trikatu treated arthritic rats which suggest the anti-inflammatory effect of the drug.

Tumor necrosis factor (TNF-α), a pro-inflammatory cytokine is a potent mediator of inflammation and it is produced primarily by the monocytes and macrophages. TNF-α is an autocrine stimulator as well as potent paracrine inducer of other inflammatory cytokines, including interleukin-1, interleukin-6, and interleukin-8. TNF-α also promotes inflammation by stimulating fibroblasts to express leukocytes adhesion molecule, resulting in increased transport of leukocytes into inflammatory sites, including the joints in rheumatoid arthritis patients [1]. Several lines of evidence support that TNF-α plays a crucial role in inflammatory arthritis [23]. In experimental arthritis, TNF-α drives early joint inflammation and the inflammatory cell infiltration in the synovial tissue. TNF-α triggers the production of other cytokines and endothelial adhesion molecules, stimulates collagenase and induces osteoclast differentiation [24]. Several studies have shown that interleukin-1 β (IL-1β), an arthritogenic polypeptide can induce many of the biological and immunological abnormalities that exist in rheumatoid arthritis including induction of acute phase proteins from the liver, release of tissue degrading enzymes and prostanoids from the synovial membrane and the resorption of bone and cartilage [25]. Hence the blockade of TNF-α might have a beneficial effect on inflammation. In our study, a significant reduction in the TNF-α and IL-1 β was found in the serum of trikatu treated arthritic rats when compared to those of untreated arthritic rats. This inhibitory effect might be due to the presence of ginger compounds, piperine (active principle of black pepper) and piperinic acid (active principle of *P. longum*) which have shown to exhibit suppressive action on pro-inflammatory cytokines produced by synoviocytes, chondrocytes, leukocytes and peripheral blood mononuclear cells [18,26,27].

Active constituents of trikatu such as piperine (active principle of black pepper), piperinic acid (active principle of *P. longum*) 6-gingerol, 6-shogaol and other ginger compounds are reported to have promising immunomodulatory properties, along with their antioxidant and anti-inflammatory activities. Therefore, the observed immunomodulatory and anti-inflammatory activity of trikatu may be attributed from these constituents [18,28–30]. In conclusion, the results of our current study indicate that trikatu may act as a therapeutic candidate for modulating the immune responses in autoimmune disease like rheumatoid arthritis. However, studies at the molecular level are required to establish the exact mechanism of action of the drug, which are underway.

Competing interest

The authors declare that they have no competing interests.

References

[1] E.H. Choy, G.S. Panayi, Cytokine pathways and joint inflammation in rheumatoid arthritis, N. Engl. J. Med. 344 (2001) 907–916.
[2] I.B. McInnes, G. Schett, Cytokines in the pathogenesis of rheumatoid arthritis, Nat. Rev. Immunol. 7 (2007) 429–442.
[3] A.O. Tzianabos, Polysaccharide immunomodulators as therapeutic agents: structural aspects and biologic function, Clin. Microbiol. Rev. 13 (2000) 523–533.
[4] A.R. Annamalai, R. Manavalan, Effects of trikatu and its individual components and piperine on gastrointestinal tract, Indian Drugs 27 (1990) 597–604.
[5] G. Ramadan, M.A. Al-Kahtani, W.M. El-Sayed, Anti-inflammatory and anti-oxidant properties of *Curcuma longa* (turmeric) versus *Zingiber officinale* (ginger) rhizomes in rat adjuvant-induced arthritis, Inflammation 34 (2011) 291–301.
[6] V. Murunikkara, S.J. Pragasam, G. Kodandaraman, E.P. Sabina, M. Rasool, Anti-inflammatory effect of piperine in adjuvant-induced arthritic rats – a biochemical approach, Inflammation 35 (2012) 1348–1356.
[7] E.P. Sabina, S. Nagar, M. Rasool, A role of piperine on monosodium urate crystal-induced inflammation – an experimental model of gouty arthritis, Inflammation 34 (2011) 184–192.

[8] M. Rasool, P. Varalakshmi, Suppressive effect of *Withania somnifera* root powder on experimental gouty arthritis: an in vivo and in vitro study, Chem. Biol. Interact. 164 (2006) 174–180.

[9] F. Benencia, M.C. Courrèges, F.C. Coulombié, In vivo and in vitro immunomodulatory activities of *Trichilia glabra* aqueous leaf extracts, J. Ethnopharmacol. 69 (2000) 199–205.

[10] D.A. Subramoniam, S. Evans, S. Rajasekharan, P. Pushpangadan, Effect of *Trichopus zeylanicus Gaertn* (active fraction) on phagocytosis by peritoneal macrophage and humoral immune responses in mice, J. Ehanopharmacol. 32 (2000) 221–225.

[11] N.K. Jerne, A.A. Nordine, Plaque forming agar by single antibody producing cells, Science 40 (1963) 405–407.

[12] R. Gonda, M. Tomoda, N. Shimizu, M. Kanari, Characterization of an acidic polysaccharide from the seeds of *Malva verticillata* stimulating the phagocytic activity of cells of the RES, Planta Med. 56 (1990) 73–76.

[13] M. Rasool, E.P. Sabina, Antiinflammatory effect of the Indian Ayurvedic herbal formulation Triphala on adjuvant-induced arthritis in mice, Phytother. Res. 21 (2007) 889–894.

[14] P. Seth, R.V. Srinivas, Circulating immune complex in cervical cancer: simple method for detection and characterization, Indian J. Med. Res. 73 (1981) 926–929.

[15] L.E. Miller, in: H.R. Ludke, J.E. Peacock, R.H. Tomar (Eds.), 2., Manual of laboratory immunology, Lea and Febiger, London, 1991, pp. 1–18.

[16] P.W. Askenase, H. Van Loverent, Delayed-type hypersensitivity: activation of mast cells by antigen-specific T-cell factors initiates the cascade of cellular interactions, Immunol. Today 4 (1983) 259–264.

[17] H. Zhou, Y. Deng, Q. Xie, The modulatory effects of the volatile oil of ginger on the cellular immune response in vitro and in vivo in mice, J. Ethnopharmacol. 105 (2006) 301–305.

[18] P. Devan, S. Bani, K.A. Suri, N.K. Satti, G.N. Qazi, Immunomodulation exhibited by piperinic acid through suppression of proinflammatory cytokines, Int. Immunopharmacol. 7 (2007) 889–899.

[19] H. Jonsson, P. Allen, S.L. Peng, Inflammatory arthritis requires Foxo3a to prevent Fas ligand–induced neutrophil apoptosis, Nat. Med. 6 (2005) 666–671.

[20] B.N. Halpern, B. Benacerraf, G. Biozzi, Quantitative study of the granulopectic activity of the reticulo-endothelial system. I. The effect of the ingredients present in India ink and of substances affecting blood clotting *in vivo* on the fate of carbon particles administered intravenously in rats, mice and rabbits, Br. J. Exp. Pathol. 34 (1953) 426–440.

[21] A.N. Theofilopoulous, F.J. Dixon, The biology and detection of immune complexes, Adv. Immunol. 28 (1979) 89–220.

[22] F.C. Hay, L.J. Nineham, R. Perumal, I.M. Roitt, Intra-articular and circulating immune complexes and antiglobulins (IgG and IgM) in rheumatoid arthritis; correlation with clinical features, Ann. Rheum. Dis. 38 (1979) 1–7.

[23] M. Feldmann, F.M. Brennan, R.N. Maini, Rheumatoid arthritis, Cell 85 (1996) 307–310.

[24] F. Bazzoni, B. Beutler, The tumor necrosis factor ligand and receptor families, N. Engl. J. Med. 334 (1996) 1717–1725.

[25] C.M. Huang, F.J. Tsai, J.Y. Wu, M.C. Wu, Interleukin-1beta and interleukin-1 receptor antagonist gene polymorphisms in rheumatoid arthritis, Scand. J. Rheumatol. 30 (2001) 225–228.

[26] P.V. Phan, A. Sohrabi, A. Polotsky, D.S. Hungerford, L. Lindmark, C.G. Frondoza, Ginger extract components suppress induction of chemokine expression in human synoviocytes, J. Altern. Complement. Med. 11 (2005) 149–154.

[27] J.S. Bang, D.H. Oh, H.M. Choi, B.J. Sur, S.J. Lim, J.Y. Kim, H.I. Yang, M.C. Yoo, D.H. Hahm, K.S. Kim, Anti-inflammatory and antiarthritic effects of piperine in human interleukin 1β-stimulated fibroblast-like synoviocytes and in rat arthritis models, Arthritis Res. Ther. 11 (2009) R49.

[28] G.S. Bae, M.S. Kim, W.S. Jung, S.W. Seo, S.W. Yun, S.G. Kim, R.K. Park, E.C. Kim, H.J. Song, S.J. Park, Inhibition of lipopolysaccharide-induced inflammatory responses by piperine, Eur. J. Pharmacol. 642 (2010) 154–162.

[29] B.H. Ali, G. Blunden, M.O. Tanira, A. Nemmar, Some phytochemical, pharmacological and toxicological properties of ginger (*Zingiber officinale* Roscoe): a review of recent research, Food Chem. Toxicol. 46 (2008) 409–420.

[30] M. Meghwal, T.K. Goshwami, *Piper nigrum* and piperine: an update (review), Phytother. Res. 27 (2013) 1121–1130.